Korea Technology and Information Promotion Agency for SMEs

(max 10 pages)

Technology Proposal for 2026 TIPA* International Joint R&D Program

(TIPA: Korea Technology and Information Promotion Agency for SMEs)

The objective of this Technology Proposal is to introduce the institute's proprietary technologies and present potential collaborative areas for Korean SMEs, facilitating joint research and development projects under the 2026. This Technology Proposal will be publicly disclosed in conjunction with the official announcement of the 2026 TIPA International Joint R&D Program.

1. Technology Proposer Information

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2. Technology Overview

Title of Proposed Technology	***Reliable Digital Twins via Scientific Machine Learning and Uncertainty Quantification***				
Korea's 12 National Strategic Technology Fields	**Semiconductor/Display (), Secondary Battery (), Advanced Mobility ( ), Next-gen Nuclear Power ( ), Advanced Bio ( ), Aerospace/Marine ( ), Hydrogen ( ), Cybersecurity ( ), AI (v), Next-gen Communication ( ), Advanced Robotics & Manufacturing (v), Quantum ( )**				
Industrial Technology Classification [1)]	**Technology Field 1** (Large Category)			**Technology Field 2** (Large Category)	
	Materials Manufacturing			***Artificial Intelligence***	
	Medium Category		Small Category	Medium Category	Small Category
	R&D / Engineering Services		***Production Process Modeling / Simulation***	***Scientific Machine Learning***	***Operator Learning***
Concept of Proposed Technology	*The proposed technology leverages **Scientific Machine Learning (SciML)** and **Derivative-Informed Neural Operators (DINO)** to create real-time, high-fidelity Digital Twins for complex engineering systems. Unlike traditional "black-box" AI, this approach integrates physical laws (PDEs) and **Uncertainty Quantification (UQ)** to provide reliable, error-controlled predictions. The final product will be a software module that accelerates engineering design and optimization processes by orders of magnitude compared to traditional FEM/CFD solvers, while providing rigorous risk assessment.*				

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Rationale for Joint R&D Project	***Industry Pain Point:*** *Small and Medium Enterprises (SMEs) in manufacturing and engineering often rely on expensive, slow commercial simulation software (e.g., FEM/CFD) for product design. This slowness bottlenecks innovation and prohibits real-time control or optimization.* ***The Opportunity:*** *While AI surrogates offer speed, industrial partners are hesitant to adopt them due to reliability concerns ("hallucinations") and lack of error estimates.* ***Synergy:*** *My research group at Georgia Tech possesses state-of-the-art algorithms for* ***scalable, reliable operator learning, optimal fluid control, shape/material design, Bayesian data assimilation, and optimal experimental design****. However, we lack the proprietary industrial data and specific real-world use cases (e.g., semiconductor manufacturing fluid dynamics, battery material design, or autonomous system control) that a Korean SME partner can provide.* ***Market Viability:*** *By combining GT's advanced algorithms with an SME's domain expertise, we can commercialize a "Next-Generation CAE" tool that allows engineers to run thousands of design iterations in minutes with confidence, addressing a critical market need in the digital transformation (Industry 4.0) sector.*
Key Words	*1. Scientific Machine Learning* *2. Predictive Digital Twins* *3. Uncertainty Quantification* *4. Real-time Engineering Simulation* *5. Optimal data acquisition and assimilation*

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3. Research and Development Objectives & Plan

Technology Readiness Level (TRL) [2)]		**Start of the Project**	Level (3)	**End of the Project**	Level (6)
R&D Objectives		*To develop a robust software framework that constructs data-efficient, physics-informed neural operators for the partner's specific engineering application. The system will demonstrate: (1) **Speed:** >100x acceleration over traditional solvers, (2) **Accuracy:** <1% relative error in key quantities of interest, and (3) **Reliability:** Accurate posterior error estimates for risk management.*			
R&D Target Specifications		- *Develop Derivative-Informed Neural Operator (DINO) architectures tailored to the partner's simulation data.* - *Benchmark accuracy against high-fidelity solvers (FEM/CFD).*			
R&D Plan		***Milestone 1 - Year 1: Offline Training & Surrogate Modeling*** • *Develop Derivative-Informed Neural Operator (DINO) architectures tailored to the partner's simulation data.* • *Benchmark accuracy against high-fidelity solvers (FEM/CFD).* ***Milestone 2 - Year 2: Uncertainty Quantification & Optimization*** • *Integrate Bayesian UQ modules to quantify predictive confidence.* • *Implement "Optimization under Uncertainty" (OUU) workflows for robust design.* ***Milestone 3 - Year 3: Software Integration & Validation*** • *Package algorithms into a deployable software module/API.* • *Validate performance on the partner's real-world hardware or production data.*			

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4. R&R (Roles and Responsibilities)

Expected R&R of Research Institute	***Core Algorithm Development:*** *• Design and train neural operator architectures (DINO, DeepONet extensions) capable of handling high-dimensional parametric dependencies.* *• Develop UQ methodologies (Bayesian inference, error estimation) to ensure model reliability.* *• Provide software libraries (PyTorch/JAX-based) for the core inference engine.*
Expected R&R of Korean SME Partner	***Domain Application & Productization:*** *• Provide high-fidelity simulation data or experimental data for training.* *• Define the specific engineering problem (e.g., heat dissipation, structural stress, fluid flow) and performance metrics.* *• Integrate the GT-developed inference engine into their existing commercial software or internal workflow.* *• Conduct validation tests in a relevant industrial environment.*

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5. Expected Outcomes/Impact

Expected Outcomes and Impact	***Commercializable Software Module:*** *A fast, reliable simulation engine that the SME can integrate into their products (e.g., CAD/CAE tools) or use to optimize their manufacturing line.* ***Cost Reduction:*** *Significant reduction in computational costs for the SME by replacing expensive high-fidelity simulations with neural operators during the design loop.* ***Enhanced Competitiveness:*** *Enabling "inverse design" and "robust optimization" capabilities that are currently computationally prohibitive for most SMEs, giving the partner a technological edge in the global market.* ***Talent Development:*** *Funding will support PhD students at Georgia Tech, fostering cross-cultural research collaboration between the US and Korea.*

6. References

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4. Qiu, Y., Bridges, N., & Chen, P. (2024). Derivative-enhanced deep operator network. *Advances in Neural Information Processing Systems*, *37*, 20945-20981.

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☐ Terms:

1) Industrial Technology Classification

· Classification system according to Industrial Technology Classification Table (**Appendix 1**, based on Article 16 of the Common Operational Guidelines for the Industrial Technology Innovation Project in Korea).

2) Technology Readiness Level (TRL)

Description		
·A consistent and objective indicator of the technological maturity of key element technologies		
	Basic Research Stage (1-2)	1st Stage: Basic theories and experiments 2nd Stage: Proof of Concept, such as practical ideas or patents
	Experimental Stage (3-4)	3rd Stage: Verification of basic performance at a laboratory scale 4th Stage: Evaluation of key performance of materials, components, and systems at a laboratory scale
	Prototype Stage (5-6)	5th Stage: Production and performance evaluation of prototypes of finalized materials, components, and systems 6th Stage: Production and performance evaluation of prototypes at a pilot scale
	Productization Stage (7-8)	7th Stage: Reliability testing and evaluation by potential customers 8th Stage: Prototype certification and standardization
	Commercialization Stage (9)	9th Stage: Commercialization

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